Hunting for signs of stable mass transfer in white dwarf binaries

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ABSTRACT

White dwarfs in binary and multiple star systems retain a record of their evolutionary history through properties such as mass, orbital separation, and the rotational behaviour of any surviving companion. This project combines two catalogues of candidate white dwarf binary systems, containing astrometric and orbital data derived from Gaia, with photometric time-series data from the Transiting Exoplanet Survey Satellite (TESS), to determine companion rotational periods and then investigate their relationship to white dwarf mass and orbital separation. Rotational periods were obtained from TESS light curves using two independent methods, the Lomb-Scargle periodogram and the Autocorrelation Function, applied with a series of statistical and consistency checks to account for the irregular sampling characteristics of TESS photometry. Of a final dataset of 349 targets, four were identified as potential close binaries or triple systems through visual inspection of their raw and phase-folded light curves. The relationship between white dwarf mass and companion rotational period reveals that all systems with a white dwarf mass below approximately $0.4-0.5M_{\odot}$ are rapidly rotating, consistent with formation via binary interaction. More unexpectedly, systems with a white dwarf mass above approximately $1M_{\odot}$ are also found to be almost universally rapidly rotating, despite their comparatively wide orbital separations. A geometric argument suggests that a significant fraction of these wide-orbit systems may be undetected triples, in which an unresolved close inner pair, rather than the wide white dwarf companion, is responsible for the observed rapid rotation.

1 INTRODUCTION

White dwarfs represent the evolutionary endpoint of the vast majority of stars, including the Sun, once nuclear fusion has ceased and the outer envelope of the progenitor star has been shed. When a white dwarf is found in a binary or multiple star system, its properties (mass, orbital separation, and the rotational behaviour of both components) provide a valuable record of the system's evolutionary history, since close interactions between stellar components can significantly alter these properties relative to what would be expected from isolated stellar evolution.

Two formation pathways are of particular relevance to interacting white dwarf binary systems. In stable mass transfer, material is transferred gradually from a companion star onto the white dwarf progenitor (or vice versa, if prior to white dwarf formation) as one star overflows its Roche lobe, without a large change in orbital separation. This process can transfer angular momentum to the accreting star, spinning up its rotation. Contrastingly in common envelope evolution, one star expands (typically during a giant phase) that its outer envelope engulfs its companion entirely. Drag forces within the shared envelope rapidly shrink the orbital separation of the two objects before the envelope is ejected, leaving behind a much closer binary than existed prior to this phase. These two pathways can therefore leave distinct observational signatures (differences in orbital separation, and potentially in the resulting rotational properties of the surviving companion) that this project aims to exploit using TESS photometric data.

In particular, the rotation of a star can be substantially affected by interaction with a companion. Isolated, single stars are expected to spin down considerably over billions of years due to magnetic braking (Skumanich 1972), such that by the time a low-mass star has evolved into a white dwarf, any surviving companion star should typically be a comparatively slow rotator. Rotation periods significantly shorter than this expectation therefore serve as a useful indicator of past or ongoing binary interaction, such as tidal locking, mass transfer, or spin-up resulting from a common envelope phase, even in systems where direct evidence of binarity, such as eclipses may not be observable due to orbital inclination.

Two space based missions are central to the work of this project, Gaia and TESS. Gaia is an astrometric survey mission from the European Space Agency (ESA) that has measured precise positions, proper motions and parallaxes for over a billion stars in the galaxy, which enables the identification of binary and multiple star systems through subtle changes in a target star's motion caused by an unseen companion (Gaia Collaboration et al. 2016). The Transiting Exoplanet Survey Satellite (TESS) is a NASA all-sky photometric survey, that was launched with the primary focus to detect exoplanets via the transiting method, which continuously monitors the brightness of stars across nearly the entire sky (Ricker et al. 2014). While originally designed for exoplanet detection, TESS's high-cadence photometric series also provides a valuable resource for studying stellar variability, including the periodic brightness variations produced by rotating, spotted stellar surfaces, which is the primary observation that this project aims to harness to determine companion rotational periods.

White dwarfs are typically many magnitudes fainter than an F, G or K type main sequence companion at optical wavelengths, meaning that such binaries cannot generally be identified though direct imaging or conventional spectroscopy techniques alone (Parsons et al. 2016). Both datasets instead rely on indirect detection signatures to identify any candidate systems. This project makes use of two datasets of candidate white dwarf binary systems. Garbutt et al. (2024) presents Gaia-derived astrometric orbits for a sample of previously known, spectroscopically confirmed white dwarfs with an F, G or K type main sequence companion, identified using an ultraviolet excess indicating the presence of a hot white dwarf beneath the light of a cooler companion star. Prior to this work, very few of these systems had measured orbital periods or component masses despite being long known to exist. This detection method is biased towards younger and hotter white dwarfs that are capable of producing a measurable UV excess, which potentially misses older and cooler white dwarfs in any otherwise similar systems.

Shahaf et al. (2024) instead constructs a substantially larger, newly-identified candidate sample by applying a statistical triage technique directly to the full Gaia DR3 catalogue of astrometric non-single star solutions. This distinguished any likely white dwarf companions from ordinary main sequence binaries or triples on the basis of the astrometric mass function and photometric colour excess alone, without prior spectroscopic confirmation. This approach is instead biased towards systems with an orbital period and mass ratio sufficient to produce measurable astrometric wobble within Gaia's observational baseline, which potentially misses both very short-period systems below Gaia's astrometric resolution and very wide systems that have not yet completed a sufficient fraction of an orbit to be characterised. It is important to note, that an error in the original code used in the Shahaf et al. (2024) paper warranted correction, which was published in Shahaf et al. (2025). The corrected values from this erratum are used throughout this project, although the original paper is referenced throughout for consistency with the original publication. These two datasets were selected as they are among the few available that provide Gaia-derived orbital periods for candidate white dwarf binary systems, which was a prerequisite for this project's approach of comparing an independently identified orbital period against a TESS-derived rotational period. Together, they provide a combined sample that spans both firmly confirmed systems and a larger population of statistically selected candidates.

More broadly, this work has relevance beyond characterising individual systems. one of the proposed ways a white dwarf can eventually explode as a Type Ia supernova is by slowly pulling material from it's binary companion until it reaches the Chandrasekhar mass (Whelan & Iben 1973). Since the white dwarf binaries studied here are examples of these exact kind of system, understanding how commonly they form, and potentially from which pathway, helps to test how plausible this explosion route can be.

To identify rotational periods from the TESS light curve data, two independent period-finding methods were employed: the Lomb-Scargle periodogram, and the Autocorrelation Function. These methods were applied in combination, alongside a series of statistical and consistency checks, in order to obtain a reliable rotational period estimate for each target while accounting for the irregular sampling and observational gaps characteristic of TESS photometric data.

This dissertation is structured as follows. Section 2 describes the data processing method used to combine, remove contaminants and then cross-match the initial Gaia datasets with TESS light curve data. Section 3 details the two period-finding methods applied to this light curve data, along with the steps taken to reconcile and validate the periods obtained from each. Section 4 presents the results of this analysis, including the relationship between white dwarf mass and rotational period for the full sample. Section 5 discusses these results in the context of existing binary and triple star evolutionary pathways, before Section 6 concludes with a summary of the key findings of the project and their implication for any future work.

2 METHODOLOGY - DATA PROCESSING

In this section, the full processing method, from the initial two datasets from Garbutt et al. (2024) and Shahaf et al. (2024), until the analysis of these datasets will be discussed.

Both the Garbutt et al. (2024) and Shahaf et al. (2024) datasets primarily identify candidate systems using a unique Gaia source identification number, without including the required positional and astrometric parameters for this project. Specifically, the Right Ascension (RA), Declination (DEC), and parallax of each target were needed in order to later cross match each system against the TESS Input Catalogue and obtain their photometric light curves. The Shahaf dataset also required the mass of the companion star to be obtained. These values were retrieved by uploading the dataset as a table (`TAP_UPLOAD.t1`) to the Gaia Archive's Table Access Protocol (TAP) service, and cross matching this table directly against the main Gaia DR3 source catalogue using the following ADQL script:

```
SELECT tab.*, gaia.ra, gaia.dec, gaia.parallax
FROM gaiadr3.gaia_source AS gaia
JOIN TAP_UPLOAD.t1 AS tab
USING(source_id)
```

This query returns all existing columns from the uploaded target table (`tab.*`), together with the RA, DEC, and parallax obtained from the corresponding entry in the Gaia DR3 source catalogue, matched using the shared Gaia source ID. The same approach was used to obtain each target's Gaia DR2 source ID and G-band magnitude, both of which were required for later stages in the data processing methodology, as detailed below.

With these positional and astrometric values obtained, it was necessary to identify and remove any potential nearby contaminating sources for each target before proceeding to light curve extraction. This step was required because TESS photometry is captured using large pixels, each covering approximately 21 arcseconds on the sky, considerably coarser than typical ground based or space based telescope imaging. As a result, multiple distinct stars can fall within, or contribute flux to, the same pixel used to construct a target's light curve. Where a nearby star lies close enough on the sky to a target, its light can become mixed with that of the intended target, diluting the resulting light curve with additional and unrelated flux, which potentially can distort or mask genuine periodic brightness variations that belong to the target star. To identify targets at risk of this contamination, the following ADQL script was used to search for any additional sources within a local area of 42 arcseconds (twice the average TESS pixel size) of each target, that were also bright enough to plausibly contribute meaningful contaminating flux:

```
SELECT gaia.source_id
FROM gaiadr3.gaia_source AS gaia
JOIN TAP_UPLOAD.t1 AS tc
      ON 1=CONTAINS(POINT( ICRS , gaia.ra, gaia.dec),
                    CIRCLE(  ICRS  , tc.ra, tc.dec,
                         42/3600.0))
WHERE gaia.phot_g_mean_mag < (tc.phot_g_mean_mag + 2)
AND gaia.source_id <> tc.source_id
```

This query works by drawing a circular area of radius 42 arcseconds (converted from arcseconds to degrees by division by 3600) around each target's sky position, using the `CONTAINS/CIRCLE/POINT` ADQL geometry functions to test whether any other Gaia source's position falls within this region. The magnitude condition (`gaia.phot_g_mean_mag < tc.phot_g_mean + 2`) restricts this to sources that are no more than two magnitudes fainter than the target itself, since a much fainter nearby source is unlikely to contribute a significant amount of contaminating flux. The final condition excludes the target's own entry from being matched against itself. With this list of potential contaminants obtained for the dataset, a Python script was created to remove any affected targets from the full sample, masking out any target with an identified nearby contaminant and outputs a 'clean' dataset for further processing.

Once the two datasets had been "cleaned", a new ADQL script was written to obtain values for the proper motion RA and DEC (`pmra, pmdec`) from the Gaia DR3 source table. These values were required to propagate each target's position from the Gaia DR3 reference epoch (2016) back to the reference epoch used by the TESS Input Catalogue (J2000), ensuring that source positions were consistent with those expected by TESS at the time of cross matching, rather than the current positions at Gaia DR3 epoch:

```
SELECT tab.*, gaia.pmra, gaia.pmdec
FROM gaiadr3.gaia_source AS gaia
JOIN TAP_UPLOAD.t1 AS tab
USING (source_id)
```

By obtaining these proper motion values, each target's position could be back propagated from the Gaia DR3 epoch (2016) to the J2000 epoch used by the TESS Input Catalogue. Using these back propagated positions, another Python script was written to cross match each target's Gaia DR2 source ID against the TESS Input Catalogue (hereafter referred to as TIC numbers), the Gaia DR2 source ID being used at this stage, rather than the DR3, as the TIC itself was originally constructed using Gaia DR2 source positions. This script also removed any duplicate matches that arose from systems with non-matching source IDs, a consequence of the expanded search radius used during the cross match process. For each resulting TIC number, a script was created to obtain all available .FITS light curve files from MAST, retrieving data from every TESS sector in which the target had been observed, rather than a single sector alone, in order to maximise the amount of data available for analysis for each system.

TESS light curve data is produced by two main pipelines. The Science Processing Operations Center (SPOC) pipeline provides high cadence (up to two minute) light curves, but only for a limited, preselected set of target stars chosen prior to each observing sector. The Quick Look Pipeline (QLP), by contrast, produces lower cadence (10-30 minute) light curves derived from full frame images, covering a much larger fraction of stars observed by TESS across each sector. Through discussion with the project supervisor, it was decided that it was not feasible to use the SPOC pipeline as initially intended, as insufficient targets in the combined sample were covered by SPOC's target list. The decision was made to use QLP light curves instead, in order to maximise the number of targets in the dataset with useable data. With all the .FITS files downloaded, a script was created to then organise and sort them into individual folders by TIC number, for ease of file management.

3 METHODOLOGY - DATA ANALYSIS

With the light curve .FITS files downloaded, the analysis of the data could begin. To analyse the datasets, two methods of identifying rotational period were completed: the Lomb-Scargle (LS) method, and the Autocorrelation Function (ACF) method.

3.1 Lomb-Scargle (LS) Method

The Lomb-Scargle (Lomb 1976; Scargle 1982) periodogram operates by fitting a sinusoidal model to given data at each trial frequency, with the quality of said fit expressed as a power value. At a given frequency (f), the model takes the form of a truncated version of the Fourier series:

$$y(t; f, \vec{\theta}) = \theta_0 + \sum_{\text{n=1}}^{\text{nterms}} [\theta_{2n-1} \sin(2\pi n f t) + \theta_{2n} \cos(2\pi n f t)] \quad (1)$$

where $\vec{\theta}$ are the fitted model parameters.

In the standard, single-term case (i.e., $N_{terms} = 1$), this reduces to a single sinusoidal fit. The periodogram power at each frequency is computed using the comparison of the residuals of this periodic model, $\chi^2(f)$, against the residuals of a constant reference model, χ^2_{ref}. The ASTROPY.TIMESERIES.LOMBSCARGLE implementation used in this project applies the standard normalisation equation:

$$P_{\text{standard}}(f) = \frac{\chi^2_{\text{ref}} - \chi^2(f)}{\chi^2_{\text{ref}}} \quad (2)$$

The significance of a given periodogram peak is quantified using the False Alarm Probability (FAP), which expresses the probability of observing a peak of a given height, under the null hypothesis that the data consists entirely of Gaussian noise with no periodic signal present. Since there is no exact formula for the probability that the periodogram's highest peak arose solely by chance, this probability must be approximated instead. The Baluev method (Baluev 2008) was used for this purpose, which uses statistical techniques for analysing extreme values to estimate a conservative (upper-bound) FAP. Specifically, this method treats the periodogram power as a continuous stochastic process across frequency, and derives an upper bound on the probability that this process would exceed a given power threshold anywhere across the searched frequency range. This approximation is valid in the case of alias free systems, and avoids the need to repeatedly recompute the periodogram on resampled data. This approximation remains reliable even when the observation times follow a semi-regular pattern, as is the case with TESS data, due to gaps from data downloads and sector transitions, which can otherwise introduce misleading peaks into the periodogram. This phenomenon was observed in a small number of targets within this project's sample.

Originally, the bootstrap method was used, as it offers a more direct estimate of the FAP. This method works by repeatedly resampling the observed light curve's flux values, while preserving the original observation times, and computing a new Lomb-Scargle periodogram for each resampled dataset. Since any genuine periodic signal is destroyed by this resampling process, the resulting periodograms represent the expected behaviour under the null hypothesis of pure noise, allowing the FAP to be estimated directly from the fraction of resampled periodograms whose peak power exceeds that of the real data. However, this method is more computationally expensive, requiring the order of $(10/P_{false}$, where P_{false} is the FAP) individual periodograms to be computed per target. Given the size of the datasets used in this project, this was found to be too time-costly, and the Baluev method was adopted instead, in order to more effectively use the project time.

The Lomb-Scargle method was used to identify periodic signals in the TESS lightcurve data, using the LOMBSCARGLE class from the ASTROPY.TIMESERIES package. Unlike a standard Fourier transform, the LS method is suited to uneven time series samples, like those found in the TESS lightcurve data, as it performs a least-squares fit with sinusoidal models rather than assuming there is uniform time sampling. Prior to computing the periodogram, each lightcurve was de-trended by fitting and subtracting a second-order polynomial from the flux as a function of time, in order to remove any long-term, non-periodic drifts in brightness that could distort or negatively affect the periodogram. This detrending step carries a potential risk. Since the polynomial is fitted directly to the observed flux, any genuine periodic signal with a period comparable to the timescale of the fitted polynomial trend may be partially absorbed by the fit and suppressed. This could potentially reduce the recovered periodogram power for targets that have genuinely long rotational periods, or in extreme cases, remove weaker long period signals from the light curve entirely. The trial frequency range (which is used instead of uniform sampling) was determined using the AUTOPOWER() method, evaluated over a range of 0.1 to 24 cycles per day, corresponding to a period search range of approximately 0.04 to 10 days. This range was chosen to match the cadence at which TESS observes (the fixed time interval between photometric measurements, set by the data pipeline used to produce each light curve) while ensuring that the baseline of each sector could be searched for periodicity, as each sector was analysed individually. For each trial frequency, the LS method computes a power value which indicates the goodness of the fit; with peaks in the resulting periodogram corresponding to the most likely frequency candidate for the object or system.

This method was used in conjunction with a 1% False Alarm Probability (FAP) threshold line (calculated using the FALSE_ALARM_LEVEL() method) which returns the periodogram power corresponding to a specific FAP. Peak frequencies whose power exceeded this 1% threshold were considered statistically significant, and those objects were considered to have real data (as they had a 99% confidence level that the signal was not random noise). Any objects that had a frequency peak lower than the FAP threshold were therefore rejected. It is worth noting that there are some limitations inherent to using this approach. The 1% FAP threshold determines whether a peak on a periodogram is statistically significant, but that significance alone does not guarantee that the identified period is correct. A highly significant peak could still correspond to an aliased signal, a harmonic of the true period, or could be a systemic artefact in the data, rather than the true periodicity of the target. Additionally, the Baluev approximation assumes the periodogram peaks are approximately independent of one another. Where trial frequencies are closely spaced or highly correlated, as can occur with irregular TESS sampling, this assumption may not hold exactly, potentially affecting the accuracy of the resulting FAP estimate. Finally, the FAP threshold assesses whether a signal is unlikely to have arisen from noise, but provides no actual information about the size of the uncertainty on the identified period itself.

3.2 Autocorrelation Function (ACF) Method

The autocorrelation function (ACF) quantifies the similarity between a time series and a time-lagged copy of itself, and is defined at a given time-lag (Δt) as:

$$ACF(\Delta t) = \frac{\lim_{T\to\infty} \frac{1}{T} \int_T y(t), y(t+\Delta t), dt}{\sigma_y^2} \tag{3}$$

where (σ_y) is the standard deviation of the time series. With this normalisation, (ACF(0)=1), and the function decays towards zero as the time series becomes uncorrelated with itself at larger lags. Where a periodic signal is present, the ACF exhibits recurring peaks at lag values corresponding to the underlying period and its integer multiples.

As TESS lightcurve data is sampled unevenly, due to gaps arising from data downloads and sector transitions, a direct calculation of the ACF is not appropriate. To address this, the discrete correlation function developed by Edelson & Krolik (Edelson & Krolik 1988) was used instead, which bins and averages the correlation between pairs of data points according to their time separation rather than requiring uniform sampling. This was implemented using the ACF_EK function from the ASTROML.TIME_SERIES package, which takes the lightcurve's time, flux, and flux error values as inputs, alongside a set of time-lag bins, and returns the corresponding autocorrelation values.

The bin width used to compute the ACF was matched to the native cadence of the data that produced each lightcurve, using a bin width of 0.1 days for SYS_RM_FLUX light curves, with a 10 minute cadence, and 0.05 days for KSPSAP_FLUX light curves, with a 2 minute cadence. SYS_RM_FLUX is a newer, intermediate QLP data product that was introduced from Sector 56 onwards, which has instrumental systematics removed but the stellar variability is left intact, and was explicitly designed to better support stellar astrophysics work. KSPSAP_FLUX is the QLP's detrended flux that was produced by applying a high pass filter, this preserves transit like signals yet suppresses longer timescale stellar variability. The maximum lag considered was set to half the time baseline of each sector, capped at 20 days, to ensure a sufficient number of point-pairs contributed to the correlation at each lag. To reduce the effect of noise on the resulting ACF and improve the reliability of peak identification, the ACF was smoothed using the GAUSSIAN_FILTER1D() function from the SCIPY.NDIMAGE package, with a smoothing width of sigma = 2. Peaks in the smoothed ACF were then identified using the FIND_PEAKS() function from the SCIPY.SIGNAL package, with a minimum prominence of 0.05 to exclude minor fluctuations not representing genuine periodicity, and a minimum peak separation of 0.5 days. The lag value of the first identified peak was taken as the candidate period for the target. Unlike the Lomb-Scargle periodogram, where multiple significant peaks were treated as indicative of an ambiguous or unreliable detection, the presence of multiple peaks in the ACF was expected and consistent with a genuine periodic signal, as peaks recur at integer multiples of the true period.

The ACF method offers a complementary approach to the Lomb-Scargle periodogram, with different strengths and weaknesses. The Lomb-Scargle method is based fundamentally on fitting a sinusoidal model to the data, meaning it can struggle to accurately identify the period of a signal with a non-sinusoidal shape. By contrast, the ACF method does not assume the shape of the targets periodic signal, and therefore is able to more reliably identify the period for non-sinusoidal signals. However, the ACF method is less well-suited for formal statistical significance testing that the Lomb-Scargle method. While a 1% FAP threshold provides the Lomb-Scargle method with a strict significance criteria, there was no direct equivalent applied to the ACF method in this project, meaning that the ACF period identifications had to rely on the peak-prominence criteria described above. Due to this reason, the two methods were used in conjunction, rather than relying solely on one method or the other.

3.3 Reconciling the Lomb-Scargle and ACF Methods

While the Lomb-Scargle periodogram and the Autocorrelation function were each applied independently to identify a target's period, several steps were taken to reconcile their outputs and assess the reliability of the period obtained.

Where the Lomb-Scargle period was rejected (due to failing the 1% FAP threshold) the ACF-derived period was used in its place for phase-folding the light curve. This ensured that a target's period remained available for further analysis wherever the ACF method had identified a clear peak, even in cases where the Lomb-Scargle result was deemed unreliable.

To assess the consistency between the two period-finding methods across the full sample, the periods obtained from each method were compared directly via a ratio plot, with the Lomb-Scargle period plotted on the x-axis against the ACF period on the y-axis for each target. Ratio lines of 1:1, 1:2 and 2:1 were plotted under the data to see if there was any correlation and if the methods agreed exactly. Targets falling along the 1:1 line were considered to show strong agreement between the two methods, lending confidence to the derived period.

Finally, for targets that were observed over multiple TESS sectors, the consistency of each method's period estimate was checked independently across sectors. For each target, the largest cluster of period estimates falling within a tolerance of 0.05 days of one another was identified, separately for the Lomb-Scargle and ACF results. Sectors whose period estimate fell within this cluster were considered to be consistent, while those falling outside of it were flagged as inconsistent. In cases where a sufficiently large cluster of consistent sectors was identified, the mean period was taken as a suggestion for the period, providing an additional and independent check on the reliability of the periods obtained by each method.

When analysing each sector, the plots produced by the python script were manually observed, to ensure that all plots displayed clear periodic light curves (both raw and phase-folded, with clearly identifiable peaks in the LS periodogram and a clear sin wave in the ACF periodogram. If these manual checks of the plots were not acceptable according to the checks described through the methodology, then that target sectors estimated period was removed from the cluster and disregarded from the consistency check. Figure 1 shows an example of a manually checked target that has met all the criteria for being a candidate system.

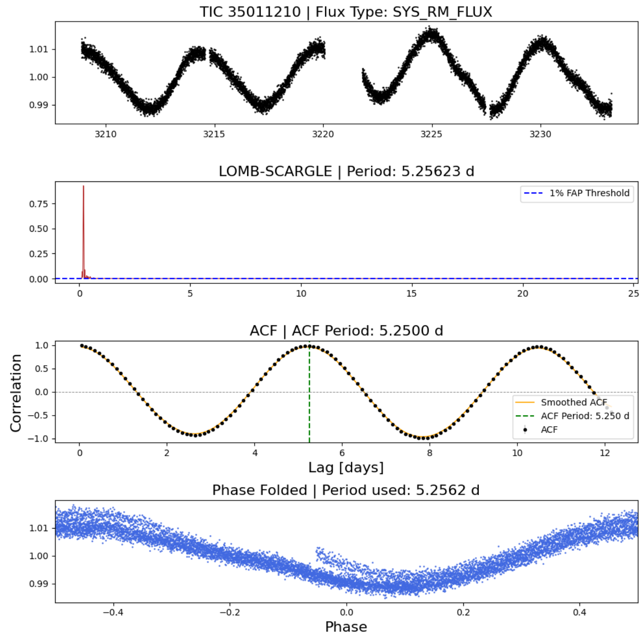

Figure 1. A four section subplot. **TOP**: Raw, normalised light curve, showing a clear periodic variability, **UPPER-MIDDLE**: Lomb-Scargle periodogram, showing a single dominant peak (red peak) at 5.25623 days, above the 1% FAP threshold line (blue dashed line). **LOWER-MIDDLE**: ACF Periodogram (black points) and the gaussian-smoothed form (orange line) showing a clear peak at a lag of 5.2500 days. **BOTTOM**: Light curve phase-folded on the Lomb-Scargle period, showing a smooth, near-sinusoidal curve.

3.4 Binary Target Identification & Analysis

Throughout the analysis of this project, several targets were identified as potential close binaries. While the same method of period estimation was completed, the light curve data of the target allowed for the determination of if it was a potential close binary or not. Figure 2 displays the same subplots as Figure 1; unlike Figure 1, the light curve data for this target is considerably noisier, with the periodic signal less visually distinct in the raw data panel. Despite this, the phase-folded light curve reveals a shallow eclipse on the left-hand side of the phase curve (see Figure 2 bottom panel, phase ≈ -0.4). This provides visual evidence of a potenial close binary system even in cases where the underlying data is comparatively poor (when compared to Figure 1).

Notably, this target's LS periodogram displayed multiple significant peaks affected by aliasing (Section 3.1), leaving the true period more ambiguous and harder to identify from the periodogram alone. Despite this, visual inspection of the phase-folded light curve still revealed evidence of an eclipse, allowing the target to be identified as a close binary. This highlights the value of visual inspection of these targets as the final step of analysis, as it provides a means of identifying genuine periodic signals even in cases where the automated methods produce inconsistent results.

This example illustrates the broader value of visual inspection as a final step in the analysis method: of the 349 targets remaining in the combined Garbutt and Shahaf dataset following the complete period-estimation and reliability-checking process described above, four targets (approximately 1.1% of the dataset) were identified as close binary candidates through this combination of automated period detection and manual light curve inspection. While this represents a small fraction of the final sample, it demonstrates that genuine periodic signals could still be recovered even from targets whose Lomb-Scargle and ACF results were otherwise inconsistent.

4 FINAL RESULTS

Following the complete data processing and period estimation method described in sections 2 and 3, a final sample of 349 targets was obtained from the combined Garbutt and Shahaf datasets, of which four were identified as a potential close binary through visual inspection of the raw and phase-folded light curves.

4.1 Consistency between LS and ACF period determination

Figure 3 shows the Lomb-Scargle period plotted against the ACF period for each target in the sample. The majority of targets, particularly those with shorter periods (below approximately four days), fall close to the 1:1 line, indicating strong agreement between the two period-determining methods across most of the sample. A subset of targets, predominantly at longer Lomb-Scargle periods (at approximately six to nine days), instead cluster along the 2:1 line, indicating cases in which the ACF method identified a period approximately twice that recovered by the Lomb-Scargle method. A smaller number of targets fall away from all three reference lines entirely, representing cases of disagreement between the two methods. The four close binary candidates identified in Section 3.4 fall close to the 1:1 line at short periods,

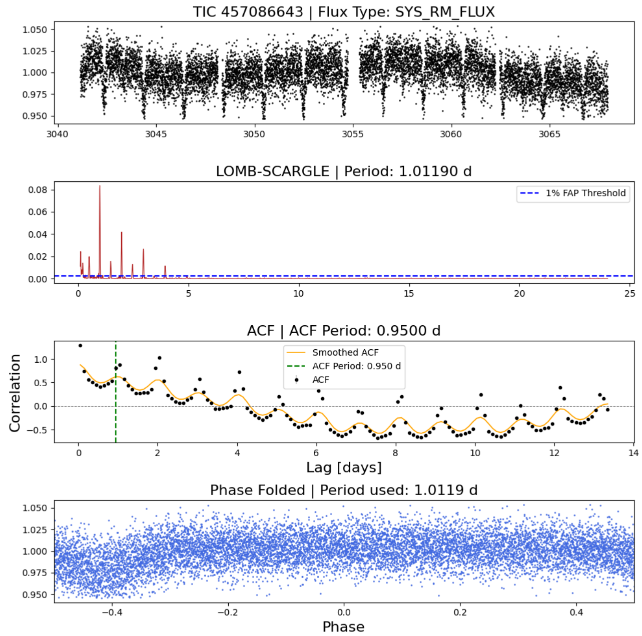

Figure 2. A four section subplot. **TOP**: Raw, normalised light curve, displaying trailing data below the main periodicity curve. **UPPER-MIDDLE**: Lomb-Scargle periodogram, showing multiple peaks that exceed the 1% FAP threshold (blue dashed line) displaying a period of 1.01190 days. This pattern is characteristic of aliasing, as discussed in 3.1. **LOWER-MIDDLE**: Autocorrelation Function, showing a weak and decaying signal, with an identified peak at a lag of 0.9500 days. **BOTTOM**: Light curve phase-folded on the Lomb-Scargle period.

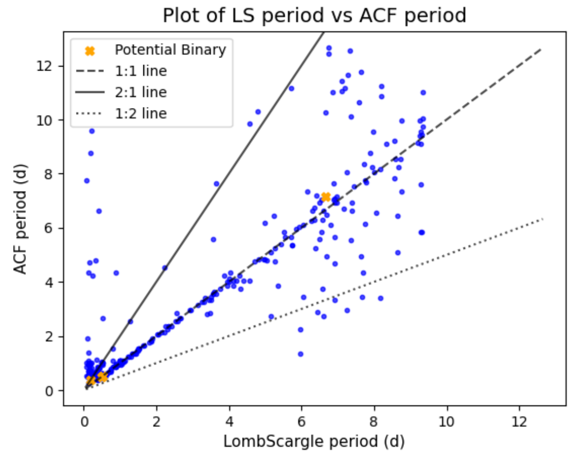

Figure 3. Plot of the Lomb-Scargle determined period against the Autocorrelation Function determined period. Also plotted are three ratio lines, displaying a 1:1, 1:2, 2:1 ratio to aid in visualising the agreement between the two methods.

indicating that both period-determining methods agreed closely for those targets, lending additional confidence to their classification.

For targets showing significant disagreement between the Lomb-Scargle and ACF periods in Figure 3, light curves were visually re-examined to assess whether their morphology was more consistent with eclipsing binary variability than star-spot variability. Of the disagreeing targets inspected, two showed light curve morphology consistent with binary eclipses. For both targets, the ACF period was found to be more consistent and reliable of an estimate, with the Lomb-Scargle period showing unclear period estimates due to limitations set out in Section 5.3. By increasing the Lomb-Scargle period of these two targets, the period agreement moved closer to the 2:1 ratio line. It should be noted that no uncertainties were calculated for either the Lomb-Scargle or ACF methods presented in this project. The degree of agreement between the two methods, discussed above, therefore provides the indication of period reliability available in this project, a limitation that is discussed further in Section 5.3.

4.2 Orbital Period vs. Rotational Period

Figure 4 shows the orbital period of each system plotted against its TESS-identified rotational period, both on a logarithmic axis, with a 1:1 ratio line shown for reference. The four close binary candidates identified in section 3.4 fall within the general trend, clustering at rotational periods of approximately 0.3-0.5 days and orbital periods of approximately 200-300 days. A small number of targets fall close to or below the 1:1 line. The physical interpretation of the relationship seen in Figure 4 is discussed in Section 5.

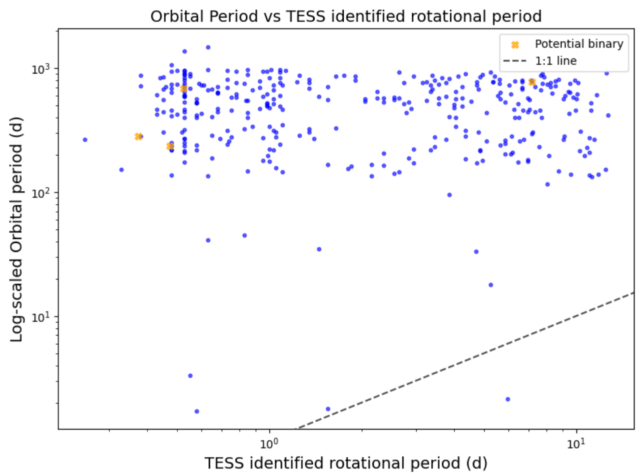

Figure 4. Plot of the TESS identified rotational period on a logarithmic scale against orbital period on a logarithmic scale. A 1:1 ratio line was plotted to determine any relationship.

4.3 White Dwarf Mass and Periastron Period

Figure 5 shows the relationship between white dwarf mass and instantaneous periastron period for the final sample, with the stable mass transfer function (Rappaport et al. 1995) and its associated boundary lines (Lin et al. 2011) shown for reference. The sample broadly follows the stable mass transfer trend, with periastron period increasing with white dwarf mass, though with considerable scatter around the reference curve. The majority of the sample, including the four close binary candidates, lies clear below the stable mass transfer curve, within the main bulk of the data points, rather than close to it. Of these candidates, at a white dwarf mass of approximately $0.85 - 1.0 M_{\odot}$, two show rapid rotation (TESS-identified periods of approximately 1-2 days), while a third shows a comparatively slow rotation period of approximately 7 days. A rotation period alone does not directly indicate a specific formation history. The possible implications of this difference are discussed further in Section 5.1. The fourth candidate lies well below the general trend, at a white dwarf mass $\approx 1.1 M_\odot$, but a substantially shorter periastron period of approximately 0.6 days. This candidate is a clear outlier with respect to both the stable mass transfer model and the wider dataset. This origin of this candidate remains unclear, and could be a candidate for future work/investigation.

5 DISCUSSION

5.1 The White Dwarf Mass-Rotational Period Relation

The relationship between white dwarf mass and TESS-identified rotational period (Figure 5) represents the central result of this project, and warrants detailed discussion. Two clear trends emerge from this plot, alongside one region of genuine ambiguity.

For systems with a white dwarf mass below approximately $0.4 - 0.5 M_\odot$, all targets but one in the sample are rapidly rotating. This is consistent with expectation, the main-sequence lifetime of a star with an initial mass low enough to produce such a low-mass white dwarf would exceed the current age of the universe by a substantial margin. As no star of this initial mass has yet had sufficient time to evolve into a white dwarf in isolation, every white dwarf in this mass range must instead have formed through some form of binary interaction (via either stable mass transfer or common envelope evolution) with the observed rapid rotation being a natural consequence of spin-up during this process. This provides strong, self-consistent evidence that the lowest-mass white dwarfs in the sample are unambiguously products of binary evolution and likely stable mass transfer.

This can be understood via the initial-final mass relation, an empirical relationship that links the mass of a white dwarf to the mass of its main-sequence progenitor. Due to substantial mass loss during the giant phases of stellar evolution, progressively lower-mass white dwarfs correspond to progressively lower-mass (and consequently longer lived) progenitor stars. Producing a white dwarf of approximately $0.3 M_\odot$ in isolation would require a progenitor of only a few tenths of a solar mass, with a main-sequence lifetime that vastly exceeds the current age of the universe, making such a formation pathway physically impossible without some form of binary interaction.

More surprisingly, systems with a white dwarf mass above approximately $1 M_\odot$ are also almost universally rapidly rotating, with only a single exception in the sample. This was not anticipated: white dwarfs of this mass are typically expected to be products of common envelope evolution occurring at wide orbital separations (with orbital periods of hundreds of days, as measured from Gaia in this sample, and shown against TESS-identified rotational period in Figure 4). There is no obvious mechanism by which such systems should have their rotation tidally locked or spun up, given the lack of close interaction implied by their wide orbits. This unexpected rapid rotation is discussed further in Section 5.2.

The intermediate mass range (approximately $0.5 - 1.0 M_\odot$) shows a more ambiguous picture, with both rapidly and slowly rotating systems present, as shown by the range of colours represented by the TESS-identified rotational period colour map, across this mass range in Figure 5. This range cannot be conclusively attributed to a singular formation channel using the data obtained in this project. It is plausible that this reflects a genuine mixture of post-common envelope and post-stable mass transfer systems, though this cannot be firmly established from the sample as a whole. Three of the close binary candidates identified in this sample (Section 4.3) fall within this mass range. However, as these candidates were confirmed by visual identification of an eclipse, they represent a small, favourably-inclined sample, rather than a representation of the wider data sample (see Section 5.2), and their individual rotational periods (two rapid, one comparatively slower) cannot alone be used as firm evidence for the relative pervasiveness of either formation pathway across the mass range as a whole. Alternatively, the ambiguity across this mass range may instead reflect an incomplete understanding of the rate at which these systems spin-down following formation, such that some rapidly rotating systems may simply be younger, post-interaction systems that have not yet had sufficient enough time to spin-down. This ambiguity may be compounded further by the sensitivity limitations of TESS photometry (discussed further in Section 5.3) which could cause genuinely spun-up, but slowly rotating, systems within this mass range to go entirely undetected, rather than appearing as a confirmed slow rotator.

5.2 Evidence for Undetected Triple Systems

A plausible explanation for the unexpectedly rapid rotation observed in wide-orbit, high-mass white dwarf systems (Section 5.1) is that a substantial fraction of the sample may consist of undetected triple systems, rather than genuine isolated binaries.

This project identified four targets showing clear, eclipse-like features in their phase-folded light curves (Section 3.4), classified here as potential close binaries or potential triples, rather than outright binaries. However, the following geometric argument suggests these eclipsing systems likely represent a small fraction of the true number of close, unresolved companions present in the sample. For a system of orbital period (P) and total mass (M_{total}), the orbital separation (a) can be obtained from Kepler's third law:

$$P^2 = \frac{4\pi^2 a^3}{G M_{total}} \tag{4}$$

An eclipse is only observed if the projected separation between the two stars, as seen from Earth, does not exceed the sum of their radii. This gives the critical inclination (i_{crit}) - the minimum inclination, or the maximum displacement from edge on orientation, at which an eclipse can still occur:

$$\cos(i_{crit}) = \frac{R_1 + R_2}{a} \tag{5}$$

For a system with an orbital period of 500 days, and two $M_\odot$, $R_\odot$ component stars, equation 4 gives an orbital separation of $a \approx 1.55$ AU and equation 5 gives a critical inclination of $i_{crit} \approx 89.66°$. This means that the orbit of a system must lie within approximately 0.34° of edge-on to produce a visible eclipse. For randomly oriented orbits, this corresponds to an eclipse probability of only:

$$p_{eclipse} = \frac{R_1 + R_2}{a} \approx 0.6\% \tag{6}$$

This estimate assumes a circular orbit and solar-radius component stars for illustrative purposes. Real systems in the sample will show some variation in orbital eccentricity and stellar radii. Given that only a small number of eclipsing systems were identified out of the full dataset of 349 targets, and that applying this eclipse probability across the sample would predict only ≈ 2 chance eclipses if every target shared this orbital configuration, this implies that a considerably larger fraction of the dataset could plausibly contain unresolved short-period companions that were not directly detected via eclipses due to unfavourable orbital inclination.

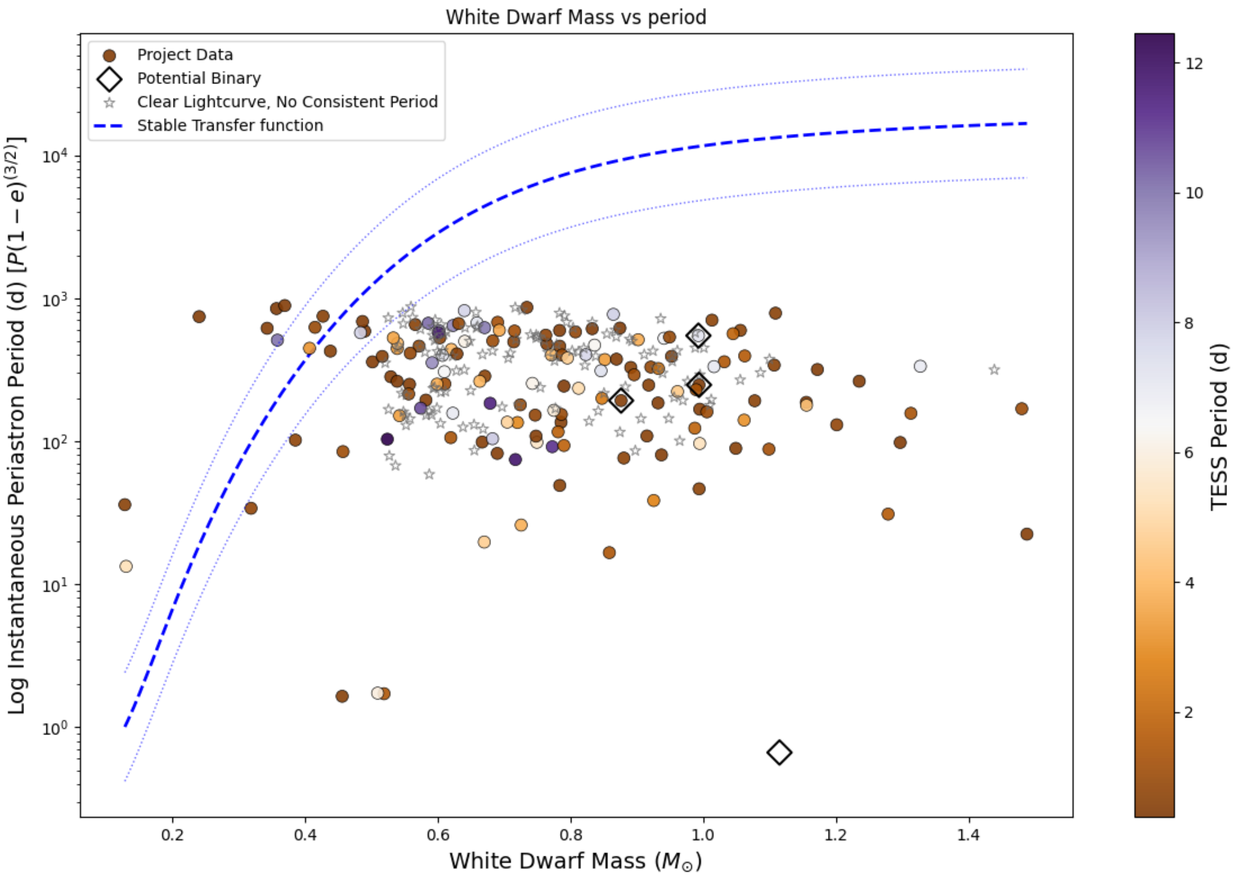

Figure 5. A plot of target White Dwarf mass ($M_\odot$) against it's instantaneous periastron period (days), as a function of $(1-e)^{3/2}$. **COLOUR MAP**: Target's TESS identified rotational period (days). Plot displays a function as seen in Garbutt et al. (2024), Rappaport et al. (1995) and Lin et al. (2011) to show if any targets lie in the conservative Stable Mass transfer window. Also plotted are the four potential binaries outlined in Section 3.4 (open diamond markers), and a series of targets, shown as open star markers, that were identified as having clear periodic light curves, but no consistent period via either analysis method.

If a significant fraction of these systems are indeed triples, this would provide a natural explanation for the rapid rotation observed even in wide-orbit, high white-dwarf mass systems: rather than being genuinely isolated binaries undergoing wide, non-interacting common envelope evolution, these systems may instead consist of an inner, closely-separated pair (potentially responsible for the observed rapid rotation via tidal interactions) orbited by the more distant white dwarf progenitor. A candidate dynamical mechanism for producing such closely separated inner pairs from an originally much wider configuration is the mass-loss induced Kozai mechanism, in which the evolution and substantial mass loss of the most massive member of a wide, hierarchical triple system disrupts the system's centre of mass, driving strong eccentricity oscillations in the remaining orbit (Kozai 1962; Lidov 1962). This mechanism has been proposed elsewhere in the literature as a means of dramatically reducing orbital separations, in some cases invoked to explain the extremely short orbital separations (Shappee & Thompson 2013) required to produce Type Ia supernovae via white dwarf-white dwarf collision (Thompson 2011). It is important to note that the current sample and methodology cannot confirm this mechanism directly; this is offered as a plausible and testable explanation for the observed trend, rather than a firm conclusion.

This interpretation would also have implications for the intermediate white dwarf mass systems that could, in principle, have evolved from isolated progenitor stars, since main-sequence lifetimes in this mass range do not exceed the age of the universe. For these systems, both an isolated-evolution pathway and a subsequent triple-induced orbital contraction remain plausible explanations for their present-day configuration, and the current data cannot distinguish between them.

5.3 Limitations

There are several limitations of the methodology used in this project that are worth noting.

Firstly, no formal uncertainty estimates were calculated for the periods obtained from the Lomb-Scargle and ACF methods. While the periods obtained from Gaia had defined uncertainties, equivalent uncertainties on the TESS-derived periods were not computed within the scope of this project, due to time constraints. Established methods do exist for this purpose, such as bootstrap resampling of the light curve. As a result, the TESS-identified periods in this project should be treated as estimates without formal error bars. The degree of agreement (or disagreement) between the Lomb-Scargle and ACF methods for a given target provides an approximate indication of the reliability of a given period estimate, since the two methods are not fully statistically independent.

Secondly, the sensitivity of a single TESS observing sector limits the range of periods that can be reliably recovered. Periods that are significantly longer than approximately 15 days become increasingly difficult to detect within a single sector's observational baseline, meaning that targets with genuinely spun-up, but comparatively slow, rotation may not have been identified as significant detections by this methodology. This introduces a potential selection bias against longer, but still anomalously fast, rotation periods within the sample.

Thirdly, despite the contaminant removal procedure applied during data processing (Section 2), the large angular size of TESS pixels means that flux from nearby bright sources may still be incorporated into the extracted light curve of a target in some cases, potentially affecting the reliability of the resulting period estimate.

Finally, a number of light curves in the sample were of insufficient quality for reliable period recovery by either method, due to noise, incomplete phase coverage, or other data artifacts. These targets were excluded from the final sample of 349 objects during the reliability-checking steps described in Section 3.3.

6 CONCLUSIONS

This project applied the Lomb-Scargle periodogram and Autocorrelation Function methods to TESS light curve data for a combined sample of white dwarf binary systems drawn from the Garbutt and Shahaf datasets, in order to identify rotational periods and assess evidence for close binarity within the sample.

Of an initial combined dataset, a final sample of 349 targets was obtained following the full data processing and reliability checking methodology described in Sections 3 and 2, of which four were identified as potential close binaries or triples via visual inspection of their raw and phase-folded light curves.

The relationship between white dwarf mass and TESS-derived rotational period (Figure 5) provides strong evidence that all white dwarfs in the sample below approximately $0.4 - 0.5 M_\odot$ formed through binary interaction, consistent with the main-sequence lifetime of their isolated progenitors exceeding the age of the universe. More unexpectedly, systems with a white dwarf mass above approximately $1 M_\odot$ were also found to be almost universally rapidly rotating, despite their comparatively wide orbital separations. A plausible explanation for this result, discussed in Section 5.2, is that a substantial fraction of the sample may consist of unresolved triple systems, in which an undetected close inner pair (rather than the wide white dwarf companion) is responsible for the observed rapid rotation; a simple geometric argument suggests the small number of eclipsing systems directly identified in this sample likely represents only a small fraction of the true number of close companions present. If confirmed by future work, this would represent a genuinely new contribution to the understanding of the formation pathways of these systems, suggesting that the wide-orbit, high white dwarf mass population may be considerably more dynamically complex than previously assumed. This has implications beyond individual system characterisation, population-level simulations of white dwarf binary evolution have typically modelled isolated binary systems in isolation from any wider dynamical environment. If a substantial fraction of wide-orbit systems are in fact triple systems, as proposed in Section 5.2, then the dynamical effects of a third body (including orbital shrinkage via mechanisms like the mass-loss induced Kozai mechanism) may need to be more broadly incorporated into such simulations to more accurately reproduce the true level of diversity of formation pathways and present day configurations observed in real white dwarf binary populations.

This project's key limitations (the absence of formal period uncertainties, the sensitivity constraints of single-sector TESS photometry, potential contamination from nearby sources, and light curve data quality) are discussed in Section 5.3, and should be addressed by any future extension of this work, alongside targeted follow-up (e.g., radial velocity monitoring) of the wide-orbit, rapidly-rotating candidates identified here to directly test the triple-system hypothesis proposed in this discussion.

REFERENCES

Baluev R. V., 2008, mnras, 385, 1279
Edelson R. A., Krolik J. H., 1988, apj, 333, 646
Gaia Collaboration et al., 2016, aap, 595, A1
Garbutt J. A., et al., 2024, mnras, 529, 4840
Kozai Y., 1962, aj, 67, 591
Lidov M. L., 1962, planss, 9, 719
Lin J., Rappaport S., Podsiadlowski P., Nelson L., Paxton B., Todorov P., 2011, apj, 732, 70
Lomb N. R., 1976, apss, 39, 447
Parsons S. G., Rebassa-Mansergas A., Schreiber M. R., Gänsicke B. T., Zorotovic M., Ren J. J., 2016, mnras, 463, 2125
Rappaport S., Podsiadlowski P., Joss P. C., Di Stefano R., Han Z., 1995, MNRAS, 273, 731
Ricker G. R., et al., 2014, Journal of Astronomical Telescopes, Instruments, and Systems, 1, 014003
Scargle J. D., 1982, apj, 263, 835
Shahaf S., Hallakoun N., Mazeh T., Ben-Ami S., Rekhi P., El-Badry K., Toonen S., 2024, mnras, 529, 3729
Shahaf S., Hallakoun N., Mazeh T., Ben-Ami S., Rekhi P., El-Badry K., Toonen S., 2025, mnras, 537, 3594
Shappee B. J., Thompson T. A., 2013, apj, 766, 64
Skumanich A., 1972, apj, 171, 565
Thompson T. A., 2011, apj, 741, 82
Whelan J., Iben Jr. I., 1973, apj, 186, 1007

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